

Week 6: Prototype Development (Phase 1)

Temporal CNN for DDoS Attack Detection

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15th February 2026

1. Goal

Run full training experiments on the BiLSTM + Attention model and the 1D Residual CNN on the BCCC-Cloud-DDoS-2024 dataset and determine whether either achieves $\geq 80\%$ 3-class test accuracy (Benign / Attack / Suspicious). The TF/Keras visual-strip baseline in `ids_2.ipynb` reaches 94.05% on binary classification but collapses to $\sim 47\%$ on 3-class due to complete Suspicious class failure. Week 6 executes the first real test of whether temporal sequence context resolves that collapse.

2. Week 5 Completion Status

The following Phase 1 deliverables were completed before proceeding with Week 6 activities:

Component	Details	Status
Data cleaning and sorting	700,774 flows parsed, timestamp-sorted, NaN/Inf $\rightarrow$ 0, protocol encoded, saved to <code>ddos_clean.parquet</code>	Complete
Feature selection	VarianceThreshold(0.001) + correlation pruning ($r > 0.97$) on first 70% rows: 318 $\rightarrow$ ~ 86 features. Selectors saved as <code>variance_selector.pkl</code> and <code>corr_keep_idx.npy</code>	Complete
Windowing	30-flow windows, stride 15, last-flow label. $\sim 46k$ windows total over full time-sorted dataset	Complete
Stratified split	70/15/15 at window level — all 3 classes guaranteed in every split. Fixes the 0-Suspicious-in-training bug from Round 1	Complete
BiLSTM + Attention	Linear($F \rightarrow 128$) + BatchNorm + BiLSTM(128, 2 layers) + soft attention + Linear($256 \rightarrow 64 \rightarrow 3$). Weights saved as <code>bilstm_ids.pt</code>	Complete
1D Residual CNN	Conv1d input proj + 2 ResBlocks ($128 \rightarrow 256$) + AdaptiveAvgPool + Linear($256 \rightarrow 64 \rightarrow 3$). Weights saved as <code>cnn1d_ids.pt</code>	Complete

Component	Details	Status
Training configuration	Adam lr=3e-4, weight decay 1e-3, label smoothing 0.1, CosineAnnealingLR, batch 256, EarlyStopping patience=10	Complete
Baseline CNN results	Binary: 94.05% accuracy, AUC 0.976. Multi-class: ~47% — Suspicious recall ~0%	Complete

3. Activities

3.1 Running Full Training Experiments

BiLSTM + Attention and 1D Residual CNN Training

- Load `ddos_clean.parquet`, reconstruct ~46k windows (30 flows, stride 15), apply the saved `temporal_scaler.pkl` and selectors, and confirm class distribution in each split matches prior runs before training begins.
- Train both models under identical conditions: same stratified splits, same random seed, Adam lr=3e-4, weight decay 1e-3, label smoothing 0.1, CosineAnnealingLR over 50 epochs, EarlyStopping on val loss (patience=10).
- Record train and val loss and accuracy per epoch for both models to detect overfitting or non-convergence as training progresses.
- Evaluate both models on the held-out test set. Primary metrics: test accuracy and macro F1. Per-class precision and recall reported separately for Benign, Attack, and Suspicious. Macro F1 is tracked alongside accuracy because a model that ignores the 8.4% Suspicious class can still reach ~91% accuracy on the majority classes — macro F1 penalises that directly.
- Generate 3×3 confusion matrices (absolute counts and row-normalised) for both models to identify which class pairs are most confused.

Comparison Against Baseline

- Compare temporal model 3-class results against the `ids_2.ipynb` visual-strip CNN (~47%, Suspicious recall ~0%) on the same label set to quantify how much sequence context improves detection.
- If either temporal model reaches ≥80%, also evaluate it on the binary task (Benign vs. Attack+Suspicious combined) and compare against the 94.05% binary baseline to check whether temporal context adds value there too.

3.2 Diagnosing Failures if the 80% Target Is Not Met

If a model does not reach 80% test accuracy, the session will diagnose the root cause before making any changes. The same failure mode that appeared in Round 1 — Suspicious windows absent from training — produced models that classified everything as Benign or Attack. Specific checks:

- Print class counts for train, val, and test splits to confirm the stratified split placed Suspicious windows in all three sets.
- Inspect val accuracy curves: if val accuracy peaks early and degrades, regularisation is insufficient; if it never improves past the ~59% benign-majority baseline, the windowing or label assignment is broken.
- Check confusion matrix Suspicious row: if Suspicious recall is near zero despite correct splits, explicit class weighting may be needed in addition to label smoothing.
- Verify that key DDoS discriminators (rst_flag_percentage_in_total, syn_flag_percentage_in_total, packet_IAT_max, packets_IAT_mean) appear in the ~86 retained features after correlation pruning.

3.3 Targeted Hyperparameter Adjustments

If diagnosis points to a fixable cause, one change will be tested at a time within Week 6. Candidates in order of likelihood to help:

Barrier	Impact	Mitigation
Suspicious class suppression despite stratified split	Suspicious recall remains ~0%; macro F1 dominated by Benign and Attack	Add inverse-frequency class weights (~3× for Suspicious) to CrossEntropyLoss alongside the existing label smoothing
Insufficient temporal context at 30-flow window size	Slower-onset Suspicious patterns from Dec 18–19 may span more than 30 flows	Test window size 50 flows (stride 25) and compare accuracy and Suspicious recall against the 30-flow baseline
Model underfitting (val accuracy consistently lower than expected but not diverging from train)	Dropout(0.4) may be too aggressive for the model size relative to the feature dimensionality	Lower dropout to 0.3 for one trial run; restore if val accuracy does not improve
Label smoothing suppressing Suspicious confidence	Smoothing at 0.1 on a 3-class imbalanced problem may prevent the model from committing to the rare class	Reduce label smoothing to 0.05 for one trial; compare Suspicious precision/recall before and after

4. Milestone / Deliverable: Prototype Training Results

4.1 Results Table

The primary deliverable is a completed training run with the following metrics documented:

Model	Test Accuracy	Macro F1	Suspicious Recall	≥80% Target?
BiLSTM + Attention	TBD	TBD	TBD	TBD
1D Residual CNN	TBD	TBD	TBD	TBD

Model	Test Accuracy	Macro F1	Suspicious Recall	≥80% Target?
CNN Visual Strip — 3-class (reference)	~47%	~0.38 est.	~0%	No

4.2 Design Decisions and Changes from Week 5

Several pipeline decisions changed between the initial Week 5 design and the current implementation. Each change is recorded with its rationale:

Decision	Week 5 Design	Current Implementation	Why It Changed
Split strategy	Temporal 70/15/15 cut on raw flows	Stratified 70/15/15 split at the window level	Plain temporal cut left only 2 Suspicious windows in training out of 32,701 total (0.006%). The test set was 38% Suspicious. Models never saw the Suspicious class and learned to ignore it entirely. Stratified window-level split guarantees all three classes appear in every split.
Scaler and selector scope	Applied separately to each of the three splits	Fit on first 70% of time-sorted rows only; applied to all windows	Fitting on all splits allowed test distribution to influence normalization and feature pruning (data leakage). Fitting on the first 70% rows only mirrors a real deployment scenario where future data is unseen at training time.
Window label assignment	Majority vote over 30 flows	Label of the last flow in the window	Majority vote silently drops Suspicious: because Suspicious is 8.4% of flows, it almost never wins a majority vote in a 30-flow window. Last-flow label preserves exact ground truth and ensures Suspicious windows are labelled correctly.
Loss function	Standard CrossEntropyLoss	CrossEntropyLoss with label smoothing 0.1	Without smoothing, the model makes overconfident predictions on the dominant classes (Benign 58.9%, Attack 32.6%) and fails to generalise to the Suspicious minority. Label smoothing softens the target distribution and forces the model to maintain non-trivial probability mass on the minority class.
Learning rate schedule	ReduceLROnPlateau	CosineAnnealingLR	ReduceLROnPlateau drops the LR sharply when validation loss plateaus, which destabilises training when combined with weight decay 1e-3 and dropout 0.4. CosineAnnealingLR decays smoothly across all epochs and is more compatible with strong regularisation.

Decision	Week 5 Design	Current Implementation	Why It Changed
Feature count	318 raw numeric features	~86 features after VarianceThreshold + correlation pruning	High-dimensional input with 318 features (many near-constant or highly correlated) produces noisy gradients and slows convergence. Pruning to ~86 features removes redundant columns while retaining the most discriminative DDoS indicators.

4.3 Supporting Outputs

- Confusion matrices for BiLSTM and 1D CNN: 3×3 absolute counts and row-normalised.
- Train and val loss and accuracy curves per epoch for both models.
- Class distribution printout confirming Benign, Attack, and Suspicious counts in each split.
- If a hyperparameter adjustment was tested: a before/after table comparing accuracy and Suspicious recall.

5. Next Steps (Weeks 7–8: Baseline Benchmarking)

Week 7 actions are conditional on Week 6 outcomes:

Week 6 Outcome	Week 7 Action
Both models ≥80% accuracy	Profile resource usage: memory footprint, training time per epoch, and inference latency per 30-flow window on Paperspace GPU. Run activity-level (26-class) classification on the best model. Evaluate both temporal models on the binary task to quantify the gain from sequence context over the 94.05% single-flow baseline.
One model ≥80%, one below	Deep diagnosis of the underperforming model. For 1D CNN: test a third ResBlock (128→256→512). For BiLSTM: test replacing the BiLSTM stack with a 2-layer Transformer encoder using scaled dot-product attention over 30 timesteps.
Both models below 80%	Re-run feature selection with tighter parameters: VarianceThreshold(0.005) and correlation cutoff $r > 0.95$. Verify that <code>rst_flag_percentage_in_total</code> , <code>syn_flag_percentage_in_total</code> , and <code>packet_IAT_max</code> are retained. Test window sizes of 50 and 100 flows to determine whether 30 flows is too short to capture the Suspicious attack signatures concentrated on Dec 18–19.